

Andrew Johnson

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Summary

I am a **Senior Data Engineer** with over 10 years of experience designing, building, and maintaining enterprise-scale data pipelines and architectures. I specialize in **ETL** development, data warehousing, and performance tuning across platforms including **Oracle, Teradata, SQL Server, Snowflake, Redshift, and BigQuery**. My expertise spans cloud-native solutions on **AWS, Azure, and GCP**, along with big data ecosystems such as **Hadoop, Spark, Hive, and Kafka**. I have a strong background in automating data workflows using **Python, Airflow, and shell scripting**, as well as implementing real-time streaming pipelines for mission-critical workloads. Skilled in **dimensional modeling, data governance, and BI tools** like **Tableau and Power BI**, I excel in delivering reliable, high-quality data solutions within Agile, cross-functional teams.

Skills

Programming: Python, Scala, R, SQL, PL/SQL, Shell Scripting, T-SQL, Bash

ETL Tools: Talend, Informatica Power Center, SSIS, Apache Airflow, Apache Oozie

Big Data & Analytics: Apache Spark, PySpark, Hive, HDFS, HBase, Cassandra, Sqoop, Apache Kafka, Databricks, AWS EMR

Cloud Platforms: AWS (S3, EMR, Glue, Lambda, Redshift, EC2, CloudWatch), Azure (Data Factory, Data Lake, Databricks, Cosmos DB), Google Cloud Platform (Compute Engine, Cloud Storage, BigQuery, Dataflow)

Databases: Oracle, SQL Server, Teradata, Greenplum,

Amazon Redshift, NoSQL (HBase, Cassandra), BigQuery

Data Warehousing: Dimensional Modeling (Star Schema, Snowflake Schema), Slowly Changing Dimensions (SCD Type 2), Data Lakes

BI & Reporting: Tableau, Power BI, SSRS

Data Formats: JSON, XML, Parquet, ORC, Avro

Workflow & Scheduling: Jenkins, Cron, Autosys, SQL Server Agent

Version Control: GitHub

Security: LDAP, Active Directory, Kerberos, IAM

Containers & DevOps: Docker, Jenkins, Puppet

Other Tools: Splunk, Postman, Alteryx, Apache Solr

Experience

SENIOR DATA ENGINEER | 07/2022 – Current

Snap Inc – Santa Monica, CA

- Architected and maintained scalable **data pipelines** for batch and streaming workloads using **AWS Glue, Apache Airflow, and Python**, supporting mission-critical data products.
- Designed and implemented **data lakehouse** architecture in **Databricks** using **Delta Lake**, structured across bronze, silver, and gold layers for governed data access and optimized performance.
- Led ingestion and transformation of high-volume, **multi-format data (JSON, XML, CSV)** from **APIs, RDBMS, and SaaS** platforms into **Snowflake, BigQuery, and Azure Synapse**.

- Developed real-time data streaming applications using **Apache Kafka**, **Spark Streaming**, and **Apache Beam**, improving data latency for reporting systems.
- Created reusable frameworks in **PySpark** and **Scala** to connect to data sources like **MySQL**, **PostgreSQL**, **Oracle**, **SQL Server**, and **Salesforce**, automating ingestion pipelines.
- Managed **ETL** orchestration using **Airflow DAGs** in **GCP Composer** and **AWS-managed workflows**, integrating with external **APIs**, **S3**, and **Redshift**.
- Built dynamic ADF pipelines with parameterized logic, event-based triggers, and **Logic Apps** for operational workflows and alerts.
- Wrote advanced **SQL** and **Python** validation scripts to reconcile datasets between landing zones, staging, and target layers with full auditability.
- Applied **SCD Type 2** logic, partitioning, and clustering strategies for slowly changing dimensions across **Snowflake** and **Hive-based models**.
- Automated deployment and monitoring of **ETL** jobs using **Docker**, **Jenkins**, and **Kubernetes**, improving pipeline reliability and reducing manual intervention.
- Integrated **REST APIs** using **Postman**, **Python requests**, and **AWS Lambda** to pull third-party data and store it in governed **S3 buckets** or raw zones.
- Delivered curated datasets and performance-tuned queries for **Power BI**, **Tableau**, and stakeholder-facing dashboards.
- Partnered with **DevOps** and **Cloud Engineering** to manage **IAM policies**, secrets, networking, and infrastructure-as-code templates (**CloudFormation/Terraform**).
- Monitored critical data jobs in **GCP Stackdriver**, **AWS CloudWatch**, and internal dashboards to ensure **SLA** adherence and real-time error handling.
- Contributed to code reviews, technical documentation, and mentoring junior engineers on data lake patterns, **ELT** strategies, and platform design.

DATA ENGINEER | 02/2017 – 06/2022

Capital One – McLean, VA

- Played a key role in the **full Software Development Life Cycle**, including **data analysis**, **modeling**, and **ETL design** across multiple enterprise platforms.
- Developed complex **ETL pipelines** using **Informatica PowerCenter**, **AWS Glue**, and **Talend**, loading data from **Oracle**, **Teradata**, and **flat files** into cloud data warehouses.
- Automated ingestion and transformation workflows using **Apache Airflow**, **PySpark**, and **shell scripts** across **AWS** and **GCP** infrastructure.
- Engineered real-time streaming jobs using **Kafka**, **Spark Streaming**, and **Scala** for ingestion into **HDFS** and **Parquet formats**.
- Worked with large-scale datasets in **HDFS**, **Hive**, **HBase**, and **Cassandra**, applying transformations and performance tuning using **Spark SQL** and **RDDs**.
- Built ingestion pipelines to migrate transactional data from on-premise systems to **Redshift**, **BigQuery**, and **Snowflake**.
- Used **Google Cloud Functions** and **Dataproc** for event-based data loading and distributed processing of raw files in cloud buckets.
- Integrated **Spring Boot microservices** for audit orchestration and data validation processes, deployed via **Oozie** and **YARN**.
- Modeled Hive tables with partitions and dynamic bucketing to optimize large-scale reporting performance.
- Designed dashboards and ad-hoc reports using **Tableau**, applying filters, parameters, sets, and dual-axis charts to support business intelligence teams.
- Created and maintained **SQL** procedures, packages, and materialized views to support real-time and batch processing.
- Wrote and tuned advanced **SQL** queries with joins, **CTEs**, window functions, and performance hints for cross-database validation and aggregation.
- Supported metadata management and data governance initiatives by integrating **Collibra APIs** with the enterprise data lake.
- Worked with DevOps tools such as **GitHub** and **Jenkins** for versioning and **CI/CD** deployment of **ETL** components.
- Collaborated with stakeholders to define requirements, review data models, and deliver scalable solutions aligned with evolving business needs.

DATA ENGINEER | 10/2015 – 01/2017

Nutanix – San Jose, CA

- Collaborated with business **stakeholders** to gather **data requirements**, define **data models**, and translate them into scalable data solutions.
- Designed and developed **ETL processes** using **SSIS** to **extract data** from **SQL Server**, **Excel**, **CSV**, and **text files** into target data marts.
- Created complex reports and dashboards with **SSRS** and **Power BI**, including parameterized, drilldown, and matrix reports from both **OLTP** and **OLAP** sources.
- Built and maintained Spark applications on **Databricks** using **Spark SQL** and **PySpark** for **data cleansing, aggregation, and transformation**.
- Migrated a legacy **Oracle-based data warehouse** to **AWS Redshift**, optimizing schema design and performance.
- Developed Spark jobs on **AWS EMR** using **Python** for processing structured and semi-structured **data** from **S3**.
- Designed **Power BI solutions**, transforming datasets with **Power Query** for data modeling and visual analytics.
- Used **AWS services** such as **EC2**, **S3**, **RDS**, and **CloudWatch** for deployment, storage, and system monitoring.
- Created and maintained shell scripts for batch processes, data loading automation, and job scheduling with **Autosys**.
- Loaded and transformed **NoSQL** datasets into **HBase** and **Cassandra** for downstream consumption.
- Tuned **SQL** and **PL/SQL** procedures for performance using execution plans, materialized views, and advanced indexing.
- Utilized **Teradata tools (FastLoad, MultiLoad, TPump)** for high-performance data ingestion and processing.
- Developed real-time and batch **ETL pipelines** in a **CI/CD environment** with **Jenkins** and **Git**.
- Implemented data quality and governance workflows using reusable **SSIS components** and **scripting**.
- Participated in **database architecture discussions, performance reviews, and unit/system integration testing**.

Education

University of Illinois Chicago – Chicago, IL | Bachelor's degree

Computer Science, 05/2015